BIF 101 (Introduction to Bioinformatics):

Mcqs for Final Term

Effort by Amina Zee



Amina Zee

1. All organisms are composed of _____.

   Cells

2. Cells are the ____unit of life.

   Basic

3. And comes from the ------ cells

   Pre-existing

4. The tendency of an atom to attract electrons towards itself is called as_____.

   Electronegativi ty

5. Living tissues are _____water

   70

6. _____ kinds of macromolecules are present.

Four

7. Macromolecules are _____ of smaller monomers.

Polymers

8. Cells are ____to maintain area to volume ratio.

Small

9. Plasma membrane _____ Traffic.

Regulate

10. Nucleoid region contain _____.

DNA

11. An octagon protein complex surrounds _____.

Nuclear pore

12. Protein fibril made _____.

Cage like structure

13. Molecular biology grew from the biochemistry and _____.

Disciplines of genetics

14. In the present age it is in the midst of _____ scientific revolution.

Two

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15. Synthesis of urea by ____ in _____.

Friedrich Wohler, 1828

16. The First enzyme was _____.

Diastase

17. Diastase was discovered by ____ in ____.

Anselme payen, 1833

18. Mendel experiment on garden pea by _____in _____.

Gregormendel, 1865

19. Chromosome theory of inheritance by____ in _____.

Thomas and Hunt morgan, 1910

20. Discovery of DNA by ___in ___.

Friedrich meisner,1869

21. Cell free fermentation by _____in ___.

Eduard Buchner,1897

22. Crystallized enzyme urease by _____in ___.

James B summer, 1926

23. Physical evidence of Recombination by ___in ___.

Barbara mac clintock and harriet crieghton, 1931

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24. One gene one enzyme was by_____in ___.

Gergor beadle and Edward tatum, 1941

25. James Watson and francis crick discover _____in ___.

Double helix structure of DNA, 1953

26. DNA is hereditary materials demonstrate by ____in____.

Oswald avery and his colleagues, 1944

27. DNA was confirmed by _____ In___.

Alfred harshey and Martha chase, 1952

28. Francis laid out the Central dogma of molecular biology in ____.

1957

29. Mathew meslson and franklin stahl proved that replication of DNA is ____

Semi -Conservative

30. Marshell nerinberg and Gobind Khorana cracked the code in _____.

1960

31. Codon stands for____

One amino acid

32. three bases constitute a code word called____

Codon

33. In 1970_____was discovered.

Gene cloning

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34. Human genome project was launched in____

1990

35. Human Genome Project completed in____

2003

36. Nuclei acid are important groups of-____

Bi molecules

37. Nuclei acid is responsible for transmission and ____

Storage

38.Nucleic acid are___ compounds.

Polymeric

39. Repeating unit of nucleic acid is____

Nucleotides

40.Nucleic acid consists of _____

RNA, DNA

41. DNA is a polymer of____

Deoxyribonucleotides

42.Deoxyribonucleotides composed of _____components.

Three

43. DNA composed of:

Phosphoric acid, nitrogenous base, deoxyribose sugar

44.A molecule containing all these three molecules is called____

Nucleotide

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45.While a molecule without the phosphate group is called____

Nucleoside

46. A Nucleotide composed of :

Nitrogenous base, sugar and Phosphoric acid

47. A Nucleoside contains :

Nitrogenous base and a sugar

48.There are ____Types of deoxyribonucleotides.

Four

49. Four types of Deoxyribonucleotides are____.

Deoxy-thymidine, Deoxy-cytidine, Deoxy-guanosine, Deoxy-aenosine

50. These four structures made up the structural unit of____

DNA

51.Succesive of nucleotides are joined together by____

Phospho diester bond

52. Base composition of DNA varies form one____to another.

Specie

53. Study of genes, hereditary and chromosomes is called___

Genetics

54. Genetic makeup of organism is called___

Genotype

55. Physical appearance of organism is called____

Phenotype

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56. First proteins sequenced in____ by ____method.

1951, Sanger and tupy

57. Atlas of protein in_____ by _____.

1960, Margret dayhoff and his collegues

58. Genome informatics is the field in which Computational and ____ techniques are applied.

Statistical

59. E-coli is ____

Bacteria

60. S- Cerevisiaeis is a:

Yeast

61. C-elegans is a ___

Worm

62. D.Melanogaster is a ____

Fly

63. Danio rerio is a ___

Zebra fish

64.Mus Musculus is ___

Mouse

65. Homo sapiens is___

You and me (Human)

66. Arabidopsis is____

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Plants

67.Prokaryotes are the organism whose genetic material is ___

Not enclosed in a nuclear membrane

68. First prokaryotic genome is _____

Hemophilus influenza

69.Prokaryotes have…. Genomes.

Simple

70. Eukaryotes have……genomes.

Larger

71. Prokaryotes have____ Genomes.

Simple

72. Eukaryotes have ____ genomes.

Larger

73. Eukaryotes have___ in their protein coding.

Introns

74. Densely packed region is ____

Heterochromatin

75. Lightly packed region____

Euchromatin

76. Densely heterochromatin ____

Take more color

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77. Lightly euchromain….

Take less color

78. Most common in children whose parents are were first cousins….

12.5%

79. Humans have over…… genes.

25000

80. These genes may produce……..different proteins.

250000

81. We need help of____ to store and recall the data.

Computers

82. Computers can ____ process this information.

Quickly

83. If you carefully search this online data with a bit of biology it is…..

Gold mine

84. Biology easily has……exciting year's problems.

500

85. Life, disease and……. are better understood.

Evolution

86. ____can be better developed after understanding molecular basis of life.

Drugs

87. _____ are systems biology are other research areas in bioinformatics.

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Protein structure, protein-protein interaction

88. Bioinformatics has enabled us to generate ____ data and its utilization.

Omic

89. Bioinformatics is relatively ____scientific area.

Young

90. Identification of hitherto unknown types are due to _____

Roles of RNA

91. Bioinformatics is full of_____

Challenges and opportunities

92. Accurate solution of protein structure is one of the____

Toughest Problem.

93. Some medicines have_____on patients.

Side effects

94. century of bioinformatics is

21st century

95. It is an era of personalized medicines and ____ of several common diseases.

Eradication

96. Genetic information flow from _____

DNA to RNA

97. All living cells ____ their DNA when they divide.

Replicate

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98. During cell division each daughter cell receive

Copy of genome.

99. ____ is the process by which two identical copies of DNA are made from an original molecule.

Replication

100. Replication occurs in the cells …… to cell division.

Prior

101. Information does not Flow

From protein to DNA or RNA.

102. Flow of information from RNA …… DNA.

To Backwards

103. When hypothesis was adopted chromosomal DNA function as a template.

In 1953

104. RNA molecules are move in _____

Cytoplasm

105. Concept of central dogma occurs in ____

1953

106. Central dogma referred by _____

Francis Crick

107. RNA chains are acts as a template for the synthesis of _____

DNA

109. Transformation of information in DNA in those molecules is termed as ;

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Central dogma

110. Central dogma is _____

DNA > RNA > Proteins

111. Proteins are used in constructing on _____

Cells

112. DNA and RNA molecules are built up from ___Types of molecules.

Four

114. Nucleotides are___

Adenine, Cytosine, uracil and guanine

115. DNA has ….

A, T, C, G

116. RNA has …

A, U, G, C

117. DNA is ___ Stranded.

Double

118. RNA is __Stranded.

Single

119. RNA helps to produce….

Proteins

120. mRNA nucleotides forms a codon of ____

Three nucleotides

121. tRNA has ___ Nucleotides.

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75 to 80

122. Conformation of tRNA allows it to bind ____

Ribosomes

123 ___ Are built up of proteins and carbohydrate.

Cells

124 Molecules are produced after ____ Of DNA into proteins.

Transformation

125. Adenosine pairs with _____

Thymine

126. Cytosine pairs with____

Guanosine

127. Which process decodes the bases in DNA.

Transcription

128. In RNA Uracil only pairs with ____

Adenosine

129. DNA gives rise to ____.

RNA

130. There are _____ Types of Ribonucleotides.

Four

131. Ribonucleotides are depending on the types of which bases.

Nitrogenous

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132. There are _____ types of ribonucleic acid.

Three

133. The ribonucleic acids are___

mRNA, tRNA, rRNA

134. mRNA is the types of RNA that carries information from____

DNA to proteins

135. Process of forming mRNA is called_____.

Transcription

136. mRNA maybe ___

Monocistronic

137. Length of mRNA is depend on ___

Length of gene

138. Which serve as adapter molecules in the process of protein synthesis?

tRNA

139. t RNA are Linked to an amino acid at one end by___

Covalently bond

140) which are components of ribosomes?

rRNA

141. rRNA is the ___Material.

Predominant

142. rRNA constitutes about ___Of ribosomes.

60%

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144. mRNA is ___stranded when it is formed from DNA.

Single

145. Conformation is mainly achieved by

Base stacking Interactions.

146. Protein coding region of mRNA is composed of non-overlapping string of codon called___

ORF

147. Translation starts at 5' and ends at

3' end

148. First and last codon of an ORF are called___

Start and stop codon

149. In bacteria start codon is usually

5'- AUG-3'

150. Eukaryotic cells always use … as start codon.

5'-AUG-3'

151. Start codon has how many functions.

Two

152. Eukaryotic mRNA almost always contain a ___

Single ORF

153. Prokaryotic cells have mRNA always contain ___

Two or more ORF

154. mRNAs containing multiple ORFs are known as ___

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Polycistronic RNAs

155) Those encoding a single ORF are known as___

Monocistronic ORF

157. The RBS typically located

3-9 Bp.

158. The phenomenon of linked translation between overlapping ORFs is known as ____

Coupling translation

159. Heart of protein synthesis is___

Translation

160. tRNA molecules are between

75 and 95 nucleotides

161. Dihydrouridine derived from …

Uridine

162. Hypoxanthine plays a role in the process of ….

Codon

163. tRNA molecules to which an amino acid is attached are said to be ___

Charged

164. tRNAs that lack an amino acid are said to be ___

Un-charged

165. Acyl linkage is___bond.

High energy

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166. Hydrolysis results in a large change in____

Free energy

168. All aminoacyl-tRNA synthetases attach an amino acid to a tRNA in how many steps.

Two

169. Two enzymatic steps are …..

Adenylation , tRNA charging

170. how many amino acids is attached to the appropriate tRNA.

20

171. Most amino acid are specified by ____

More than one Codon

172. Which Is the macromolecular machine that directs the synthesis of proteins.

Ribosomes

173. Ribosome is larger and more ___ Than the minimal machinery.

Complex

174. Polymerizing amino acid is composed of at-least ….

Three amino acids

176. Translation takes place at a rate of …….. Amino acid per second.

Only two to 20

177. In prokaryotes transcription and translational machinery are located in ___

Same Compartment

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178. Ribosomes catalyzes a _____.

Single Chemical reaction

179)Ribosomes is composed of

Two Subassemblies

180. Subassemblies of RNA are known as ____

Large and small units

181. Large subunit of RNA contain …..

Peptide transferase center

182. Peptidyl responsible for the ___

Formation of peptides bond

183. Small units of RNA contain …

Decoding center

184. Small and large units of RNA are buried within the intact of …..

Ribosomes

185. A-site is the binding site for ……

Aminoacylated-tRNA

186. P-site is the binding site for …..

peptidyl tRNA

187. Stop codons are recognized by proteins called___

Release factors

189) which process involves coding of proteins by RNA at ribosomes.

Translation

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190. Sequence of amino acid are linked by ….

Peptide bond

191. Proteins are polymers of….

Amino- acids

192. Proteins are range in size from …..

Small to Large

193. Twenty different proteins are called …..

Standard proteins.

194. Each type of protein has a____ amino acid sequence.

Unique

195) Linus Pauling and Robert Corey carefully analyzed the____

Peptide bond

196. Six atoms of peptide group are ____

Planar

197) In principle, ϕ and ψ can have any value between….

+180, -180

198. Secondary structure of protein refers to ___

Local conformation

199. Backbone of polypeptide chain extend into ____

Zig- zag Structure

200. Three dimensional arrangements of atoms of proteins are___

Tertiary structures

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201. A multi-subunit protein is also referred to as a___

Multimer

202) A multimer with just a few subunits is called_____

Oligomer

203. A single subunit is called___

Protomer

204. Oligomers can have either____Symmetry.

Rotational, Helical

205. The process of inexact matching while keeping in view the conserved residues is called sequence……

Alignment

206. Pairwise sequence alignment helps to compare _____.

Two Sequence

207. Gaps are not counted in:

Identity

208. Multiple sequence alignment involves comparison of:

Three or more sequences.

209. Pairwise alignment either local or ___

Global

210. MSA can help to align _____

Multiple Sequences

211. Progressive alignment can help to perform ___

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MSA

212) CLUSTAL is an ____

Online tool

213. Pariwise alignment is the alignment of ___

Two sequences

214. CLUSTAL is an online tool to perform _____

MSA

215. BLAST developed in _____

1990

216. BLAST is an basic _____

Local alignment tool

217. BLAST consist of how many types:

Two

218. Two main types of BLAST are ___

Proteins, nucleotides

219. Smith waterman perform ___

Local Alignment

220. Needleman waterman calculates____

Global alignment

221. FASTA developed in which year ____

1998

221. First Genebank developed in ____

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1982

222. Genebank is now under

NCBI

223. EMBL developed in ____

1980

224. DDBJ eastablised in _____

1984

225. First protein sequenced was collected in_____

1951

226. Protein sequence was collected by_____

Sanger and Tupy method

227. Atlas of protein sequenced by

Margrett and DAYhoff in 1960

228. Swiss prot is collaboration between _____

EBI and SIB

229. GEO is a____

Public repository

230. Informatics in health care is called ___

Health informatics

Best of Luck

☺Thank you ☺

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